

# Jace Kasen

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## EDUCATION

**Northeastern University**  
M.S. in Computer Science

Expected June 2028  
Vancouver, BC

**University of California, Los Angeles (UCLA)**

B.S. in Computer Science | Minor in Anthropology | GPA: 3.5/4.0

June 2026

Los Angeles, CA

*Relevant Coursework:* Algorithms & Complexity, Operating Systems, Distributed Systems, Computer Networks, Data Management Systems, Compiler Construction, Software Engineering

## TECHNICAL SKILLS

**Programming:** C/C++, Python, TypeScript, JavaScript, SQL, Bash

**Frameworks & Tools:** Next.js, React, React Native, Expo, Polars, Pandas, Node.js, Deno, Git, Docker

**Data & Databases:** PostgreSQL, Supabase (Edge Functions, RLS), Snappy Parquet, SQLite, pgTAP

**Languages:** English, Russian, Kazakh

## PROJECTS

**Counter-Strike 2 Analytics & ETL Pipeline** | [jacekasen.com/projects/cs2](https://jacekasen.com/projects/cs2)

Independent Project

*Python, Polars, Next.js, Snappy Parquet, SQLite, Supabase*

- Engineered an automated ETL pipeline ingesting 1,787 matches across 65 tournaments, reducing >380 GB of raw match replays into a compact 472 MB partitioned Parquet datastore (>99.8% storage reduction)
- Extracted and normalized 598,000+ combat events using Polars, computing 2D radar spatial projections and implementing 3-second sliding-window algorithms for trade-kill attribution
- Built a resilient ingestion crawler with browser TLS impersonation, request jitter, circuit-breaker error handling, and SQLite WAL mode to track match processing state
- Shipped an interactive Next.js dashboard featuring server-rendered tournament analytics, Canvas-based player form curves, and rating comparisons

**Flight Tracker Mobile App** | [github.com/jacekasen/flight-tracker](https://github.com/jacekasen/flight-tracker)

Independent Project

*Expo, React Native, TypeScript, PostgreSQL, Supabase, Deno*

- Built a cross-platform flight-history app for iOS, Android, and web with authenticated search, private travel history, and custom route maps with yearly recaps
- Secured flight lookups behind a Supabase Edge Function with JWT validation, persistent HMAC-hashed IP and per-user rate limits, and 15-minute cached provider responses
- Enforced Postgres Row-Level Security with cascading account deletion, normalized data against a 4,134-airport dataset, and calculated Great Circle flight distances
- Automated CI/CD workflows using GitHub Actions to run 35 Vitest unit tests, Edge Function HTTP tests, and database authorization assertions with pgTAP

**NBA Player Analytics & Salary Cap Explorer** | [jacekasen.com/projects/nba](https://jacekasen.com/projects/nba)

Independent Project

*Python, Next.js, TypeScript, PostgreSQL, Supabase, scikit-learn*

- Built a full-stack analytics platform that processes 36,000+ player-season records across 80 seasons, serving interactive career charts, peak-age curves, and forecasts via Next.js and Supabase
- Developed a historical salary pipeline normalizing contracts into Cap Share percentages across eras, supporting stacked payroll-against-cap and roster composition visualizations
- Built a leakage-safe ML pipeline with strict chronological validation splits (1976–2018 train, 2019–2022 validation, 2023–2024 test), resetting rolling features on non-consecutive seasons
- Built an automated prediction service integrating two-stage classification and regression models, correcting for survivorship bias and publishing calibrated career forecasts to Supabase

## LEADERSHIP & COMMUNITY

**President, Russian-Speaking Student Association, UCLA**

May 2024 – June 2026

- Rebuilt the organization to 80+ members, including 30–40 active participants
- Managed budgets, event logistics, and outreach for cultural programming
- Co-founded the Central Asian Student Union at UCLA